DATABASE MANAGEMENT LAB

LAB Sheet 4

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

K.R Mangalam University

Submitted by:

Deepak

(2401010085)

Submitted to

Ms. Mansi Kajal

Department of Computer Science and Engineering

School of Engineering and Technology

K.R Mangalam University, Gurugram- 122001, India

Batch – 2024-2028

Lab Assignment: SQL User Management & Access Control

Title:

Implementation of User Creation, Deletion, and Privilege Management in SQL

Problem Statement

Design and execute SQL commands to create users, manage their access privileges, and remove users in a database system. This assignment focuses on database security and access control mechanisms.

Tasks

Part A: User Creation

1. **Create a new user named user1 with a password.**
2. **Create another user user2 with limited access.**

Part B: Granting Privileges

3. **Grant SELECT and INSERT privileges on a table Student to user1.**
4. **Grant ALL privileges on a table Course to user2.**
5. **Provide WITH GRANT OPTION to user1 and observe its effect.**

Part C: Revoking Privileges

6. **Revoke INSERT permission from user1 on the Student table.**
7. **Revoke all privileges from user2 on the Course table.**

Part D: User Deletion

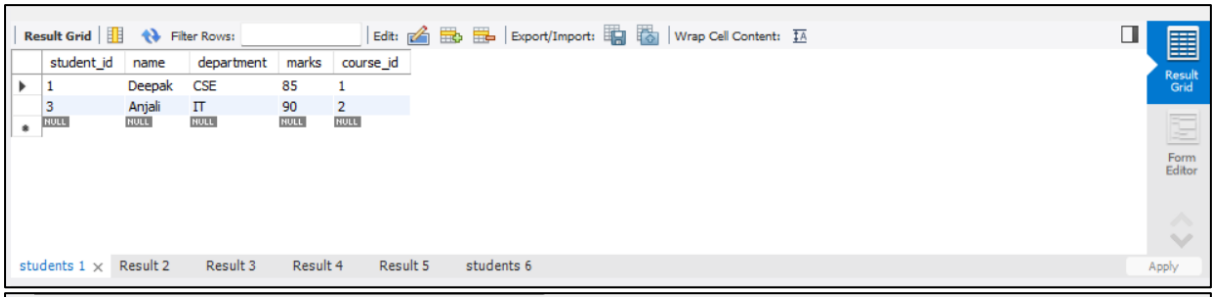
8. **Drop/Delete user user2 from the database.**
9. **Attempt to login using deleted credentials and observe the result.**

Part E: Additional Commands

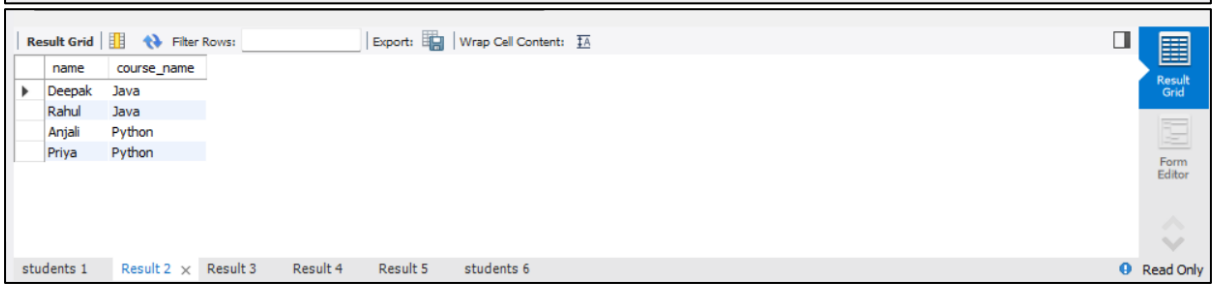
10. **Display all users in the system.**
11. **Show privileges assigned to a specific user.**

```sql
CREATE DATABASE IF NOT EXISTS college_db;
USE college_db;

CREATE TABLE students (
   student_id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   department VARCHAR(50),
   marks INT,
   course_id INT
);

CREATE TABLE course (
   course_id INT PRIMARY KEY,
   course_name VARCHAR(50)
);

CREATE TABLE employees (
   emp_id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   dept_id INT,
   salary INT
);

CREATE TABLE departments (
   dept_id INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

CREATE TABLE orders (
   order_id INT PRIMARY KEY AUTO_INCREMENT,
   customer_id INT,
   amount INT
);

INSERT INTO students (name, department, marks, course_id) VALUES
('Deepak', 'CSE', 85, 1),
('Rahul', 'CSE', 70, 1),
('Anjali', 'IT', 90, 2),
('Priya', 'IT', 60, 2);

INSERT INTO course VALUES
(1, 'Java'),
(2, 'Python');

INSERT INTO employees (name, dept_id, salary) VALUES
('Bhawna', 1, 50000),
('Aman', 2, 60000);

INSERT INTO departments VALUES
(1, 'HR'),
(2, 'IT');

INSERT INTO orders (customer_id, amount) VALUES
(1, 1000),
(1, 2000),
(2, 1500);

SELECT * FROM students
WHERE marks > 75;

SELECT students.name, course.course_name
FROM students
JOIN course
ON students.course_id = course.course_id;

SELECT department, COUNT(*)
FROM students
```

```
GROUP BY department;

SELECT e.name
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'HR';

SELECT customer_id, SUM(amount)
FROM orders
GROUP BY customer_id;

SELECT name, marks
FROM students
ORDER BY marks DESC
LIMIT 3;

START TRANSACTION;

UPDATE employees
SET salary = salary + 5000
WHERE emp_id = 1;

-- Try COMMIT or ROLLBACK
COMMIT;
-- OR
ROLLBACK;
```



student_id	name	department	marks	course_id
1	Deepak	CSE	85	1
3	Anjali	IT	90	2
NULL	NULL	NULL	NULL	NULL



name	course_name
Deepak	Java
Rahul	Java
Anjali	Python
Priya	Python

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department	COUNT(*)
CSE	2
IT	2

students 1 | Result 2 | Result 3 | Result 4 | Result 5 | students 6 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name
Bhawna

students 1 | Result 2 | Result 3 | Result 4 | Result 5 | students 6 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	SUM(amount)
1	3000
2	1500

students 1 | Result 2 | Result 3 | Result 4 | Result 5 | students 6 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	marks
Anjali	90
Deepak	85
Rahul	70

students 1 | Result 2 | Result 3 | Result 4 | Result 5 | students 6 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:33:30	CREATE DATABASE college_db	1 row(s) affected	0.031 sec
2	15:33:30	USE college_db	0 row(s) affected	0.000 sec
3	15:35:34	CREATE DATABASE college_db	Error Code: 1007. Can't create database 'college_db'; database exists	0.000 sec
4	15:35:46	CREATE DATABASE college_db	Error Code: 1007. Can't create database 'college_db'; database exists	0.000 sec
5	15:37:06	DROP TABLE college_db	Error Code: 1051. Unknown table 'college_db.college_db'	0.047 sec
6	15:37:42	CREATE DATABASE IF NOT EXISTS college_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'college_db'; database exists	0.046 sec
7	15:37:42	USE college_db	0 row(s) affected	0.000 sec
8	15:37:42	CREATE TABLE students (student_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(50), ...	0 row(s) affected	0.094 sec
9	15:37:42	CREATE TABLE course (course_id INT PRIMARY KEY, course_name VARCHAR(50))	0 row(s) affected	0.047 sec
10	15:37:42	CREATE TABLE employees (emp_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(50), d...	0 row(s) affected	0.031 sec
11	15:37:42	CREATE TABLE departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50))	0 row(s) affected	0.031 sec
12	15:37:42	CREATE TABLE orders (order_id INT PRIMARY KEY AUTO_INCREMENT, customer_id INT, amount I...	0 row(s) affected	0.031 sec
13	15:37:42	INSERT INTO students (name, department, marks, course_id) VALUES ('Deepak', 'CSE', 85, 1), ('Rahul', 'CSE', ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec
14	15:37:42	INSERT INTO course VALUES (1, 'Java'), (2, 'Python')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.016 sec
15	15:37:42	INSERT INTO employees (name, dept_id, salary) VALUES ('Bhawna', 1, 50000), ('Aman', 2, 60000)	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
16	15:37:42	INSERT INTO departments VALUES (1, 'HR'), (2, 'IT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
17	15:37:42	INSERT INTO orders (customer_id, amount) VALUES (1, 1000), (1, 2000), (2, 1500)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
18	15:37:42	SELECT * FROM students WHERE marks > 75	2 row(s) returned	0.000 sec / 0.000 sec
19	15:37:42	SELECT students.name, course.course_name FROM students JOIN course ON students.course_id = course.co...	4 row(s) returned	0.000 sec / 0.000 sec
20	15:37:42	SELECT department, COUNT(*) FROM students GROUP BY department	2 row(s) returned	0.000 sec / 0.000 sec
21	15:37:42	SELECT e.name FROM employees e JOIN departments d ON e.dept_id = d.dept_id WHERE d.dept_name = 'HR'	1 row(s) returned	0.000 sec / 0.000 sec
22	15:37:42	SELECT customer_id, SUM(amount) FROM orders GROUP BY customer_id	2 row(s) returned	0.000 sec / 0.000 sec
23	15:37:42	SELECT name, marks FROM students ORDER BY marks DESC LIMIT 3	3 row(s) returned	0.000 sec / 0.000 sec
24	15:37:42	START TRANSACTION	0 row(s) affected	0.000 sec
25	15:37:42	UPDATE employees SET salary = salary + 5000 WHERE emp_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
26	15:37:42	COMMIT	0 row(s) affected	0.000 sec
27	15:37:42	ROLLBACK	0 row(s) affected	0.000 sec